

ET 502: Embedded Systems and IoT for E-Transportation

Course Code	ET 502
Course Name	Embedded Systems and IoT for E-Transportation
L-T-P-C	3-0-2-4
Prerequisites	Digital System Design (EE 210) or equivalent
Intended for	PG
Distribution	Core for M.Tech in Electric Transportation students, elective for other advanced B.Tech /M.Tech students
Mutual Exclusion	None
Approval	44th BoA (Nov 2021)

Course Contents

- **Introduction to embedded systems:** Understanding an embedded system, design metrics, design challenges, technologies for embedded systems. (2 hours)
- **Custom Single Purpose Processor for Embedded Systems:** Design of data-paths and controllers, finite state machines, custom single purpose processor design at RT level, optimizing custom single purpose processors, introduction to hardware description languages, modeling of custom single purpose processors using hardware description languages. (3 hours)
- **Introduction to FPGA:** Introduction to complex digital systems design, notion of programmable logic devices, overview of FPGA architecture, realization of data-path and controller, timing analysis of data-path and controller, synthesis, placement, routing, performance optimization. (2 hours)
- **Introduction to Microcontrollers:** Introduction to microcontrollers, overview of architecture of a typical microcontroller such as AVR microcontroller, addressing, assembly language programming, Memory and I/O interfacing, device drivers for I/O devices. (4 hours)
- **Sensors and Actuators:** Basic principles of sensors and actuators, classification of sensors and actuators, interface methodology and circuits, integration aspects (3 hours)
- **Embedded Systems for Electric Transportation:** Battery management system (BMS), cell monitoring, battery safety and protection, state of charge estimation, state of health estimation, cell balancing, thermal management, charging control, BMS architectures. (8 hours)
- **Introduction to IoT:** Overview of Internet of Things, IoT architecture, Communication protocols (4 hours)
- **Protocols for Wired communication:** Device configuration and protocols, e.g., CAN, LIN, FlexRay, MOST, Ethernet, OBDII, (5 hours)
- **Protocols for Wireless communication:** Wifi, ZigBee, Bluetooth Low Energy (BLE) (5 hours)
- **IoT for Electric Transportation:** Centralized charging scheme, decentralized charging scheme, performance comparison and evaluation. (6 hours)

Experiments for lab: 28 hours

- Hardware modeling using hardware description language
- Modeling a custom single purpose processor for electric vehicle applications
- Hardware realization using FPGA
- Interrupt handling through microcontroller
- Stepper Motor control using microcontroller
- Battery management using microcontroller
- FPGA based battery management architectures
- Designing a multiprotocol system
- Wired communication using CAN bus, LIN, FlexRay, MOST, Ethernet, OBDII 10. TCP-UDP client server systems
- IoT application layer protocols - MQTT-CoAP
- Applications of IoT on centralized charging scheme
- Applications of IoT in decentralized charging scheme

Text Books

1. Rui Xiaong, Weixiang Shen, Advanced Battery Management Technologies for Electric Vehicles, Wiley publishers, 2019.
2. James K. Peckol, Embedded Systems: A Contemporary Design Tool, Wiley publishers, 2009.

References

1. Peter Xiao, **Designing Embedded Systems and the Internet of Things (IoT) with the ARM mbed**, Wiley publishers, 2018.
2. Edward Ashford Lee and Sanjit Arunkumar Seshia, **Introduction to Embedded Systems – A Cyber-Physical Systems Approach**, MIT Press, 2017.
3. Charles H. Roth Jr., Lizy Kurian John, **Digital Systems Design Using VHDL**, Cengage Learning, Third Edition, 2016.